

A Computational Approach to Identifying Evolutionarily Stable Strategies in Mammalian Search Behaviour

Supplementary Material A: ODD Protocol

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1 The model description follows the ODD (Overview, Design concepts, Details) protocol for
2 describing individual- and agent-based models described by Grimm et al., 2006, as updated
3 by Grimm, 2020; Grimm et al., 2010. A visual description of the model is presented in
4 Figure 5. The model was implemented in RStudio using R version 4.5.2 and model code is
5 available via online repository on [GitHub](#).

1 Purpose and patterns

The purpose of this model is to determine how mammalian search strategies, quantified by ballistic length scale, evolve and stabilise over evolutionary timescales. The ballistic length scale is defined as a function of an individuals' spatial variance in movement and their positional and velocity autocorrelation timescales. The model is designed to explore how variation in environmental factors (e.g., resource abundance and spatial distribution) and biological characteristics (e.g., body size) influence the evolutionarily stable expression of movement behaviour. By operating at evolutionary timescales and integrating energetic constraints, the model identifies the conditions under which different movement strategies are favoured and examines how optimal ballistic length scale scales with mammalian body mass across environmental conditions, expanding on encounter theory basic principles to bring further understanding in ecological contexts.

The model is evaluated against general empirical and theoretical expectations in movement ecology and physics that arise from energetic and search-related processes. These include: (1) increased ballistic length scales increases encounter rates, (2) as resource availability increases in the landscape, ballistic length scale decreases, and (3) stable movement strategies emerge.

2 Entities, state variables, and scales

The model is composed of two entities, the animal and the landscape. The animal entity represents an individual herbivorous mammal, characterised by four state variables, i) body mass (g), ii) velocity autocorrelation timescale (τ_v), iii) positional autocorrelation timescale (τ_p), and iv) spatial variance in position (σ_p). The parameters τ_v and τ_p describe the temporal persistence of velocity and position in the movement process, respectively. The parameter σ_p represents the spatial variance in position and determines the home range size of the individual and remains fixed across generations. The other movement parameters (τ_v and

τ_p) vary across generation through inheritance and mutation, while remaining fixed within
a generation. Body mass remains fixed throughout the simulation. All other animal entity
processes are described through submodels, informed through allometry (see table 3 for all
allometric equations). The landscape represents a spatial field of resource patches with which
animals interact during movement. The landscape is characterised by the spatial coordinates
 (x, y) of resource patches and the caloric value associated with each patch (*kcal*). Resource
patches are distributed across the landscape using a clustered spatial point process. The
spatial extent is defined as a function of the positional variance (see section 7.1).

The model operates at generational timescales, with evolutionary change occurring across
generations. The generation is set for a duration of one one-thousandth of the allometri-
cally determined lifespan. Within each generation, animal movement is simulated at a finer
temporal scale using a sampling interval defined as one-tenth of the velocity autocorrelation
timescale.

Table 1: State variable names, units, and variable description for animal agents.

State Variable	Description	Units
<u>Animals</u>		
Mass	the mass of individuals determined at initialization	g
τ_p	the positional autocorrelation of the animal	s
τ_v	the velocity autocorrelation of the animal	s
σ_p	the spatial variance in position	m
<u>Landscape</u>		
Patch coordinates	spatial location of resource patches	m
Patch caloric value	energy content of patch	kcal

3 Process overview and scheduling

Each generation, individuals undergo five key processes: movement, resource acquisition, energetic calculations, reproduction, and selection. Movement trajectories are generated without feedback from the landscape properties. Movement is generated within a generation in discrete time steps. Resource acquisition is determined by encounters between the movement trajectory and resource patches. Once a patch is encountered it is considered consumed and cannot be encountered again within that individual's simulation. There is no interaction between individuals such that there is no competition for resources between individuals. The landscape is reset at the beginning of each generation, and the same landscape is used for all individuals for all generations within a single trial. Net energy is determined from resource intake minus energetic costs. Individuals reproduce if net energy exceeds a reproduction threshold. Movement parameters of reproducing individuals are inherited by the next generation. The number of animal entities remains constant throughout the simulation. The resource landscape is reset at the beginning of each generation.

All processes occur synchronously across individuals, and all individuals complete each step before the next step begins. The entire process is repeated until the number of generations desired is reached. The processes proceed in the following order:

1. Initial spatial positions of individuals is assigned
2. Movement parameters are assigned to each individual
3. Movement trajectories are simulated over t time steps
4. Resource acquisition is calculated from encountered resources
5. Energy expenditures are subtracted from total gains
6. Reproductive success is determined
7. Movement parameters of successful individuals are stored to form the next generation

4 Design concepts

4.1 Basic Principles

This model is based on principles of encounter theory, in which movement determines resource encounter rates. It is built around the understanding that increased direction persistence in movement yields higher encounters with features of the landscape Visser and Kiørboe, 2006. Energy budgets are based on the metabolic theory of ecology (Brown et al., 2004), such that individual body mass informs metabolic and movement related energy expenditures. The model examines evolutionary change in movement strategies driven by differential reproductive success across generations, allowing for insight into potentially stable strategies.

4.2 Emergence

The gradual change of ballistic length scale emerges as a result of varying movement parameters over evolutionary timescales. Driving the evolution of ballistic length scale is differential fitness as a result of varying success of movement strategies. In addition, population-level distributions of movement parameters, evolutionary trajectories of ballistic length scale, and correlations between movement traits and energetic performance emerge from individual-level movement, energy acquisition, and threshold-based reproduction. No global optimization or target trait value is imposed.

4.3 Adaptation

Adaptation occurs indirectly through differential reproductive success, whereby movement parameter combinations that produce higher energetic gains contribute disproportionately to the next generation.

4.4 Objectives

Individuals do not possess explicit objectives or optimization rules. Reproductive success arises mechanistically when net energy exceeds a reproduction threshold.

4.5 Learning

Entities do not learn in the model.

4.6 Prediction

Entities cannot predict features of the model.

4.7 Sensing

There is no sensing in the model.

4.8 Interaction

Individuals are able to interact with the landscape such that they can consume and deplete resources. Resource depletion occurs within an individual's simulation only and does not alter resource availability for other individuals. Individuals are not able to alter landscapes for subsequent generations. Additionally, individuals are not able to interact with other individuals in the landscape and as such, there is no resource competition.

4.9 Stochasticity

Stochasticity is incorporated into both the animal and landscape entities. The landscape is generated from a stochastic Matérn clustering process, producing independent realisations with consistent statistical properties. Stochasticity is also incorporated into many components of the animal entities, including the initial placement in the landscape, variation in the movement parameters and the corresponding trajectories, as well as in mutation based variance in trait inheritance.

4.10 Collectives

The model does not have collective entities.

4.11 Observations

Model outputs include individual-level movement parameters, energetic gains and costs, reproductive success, and body mass. Population-level summaries of trait distributions and variances are recorded for each generation.

5 Initialization

5.1 Landscape

For each simulation, a customized landscape is generated, utilizing a Matérn cluster point process, following the landscape submodel (see section 7.1). The extent of each landscape is defined as a function of the individuals positional variance, utilizing allometric equations (see section 7.3). Each individual is supplied with the same landscape, and each individual only affects their own landscape. Consumption is destructive, such that once an individual consumes a patch, it cannot be consumed again. There is no interactions between individuals in a generation. Each generation begins with the same landscape (i.e., all resources are renewed at the beginning of each generation). For more details, view section 7.1.

The characteristics of the landscape are determined by three main parameters, i) `n_points`, ii) `mu`, and iii) `var`.

Table 2: Environmental Scenarios Tested

Scenario Component	Parameter Description	Values Tested	Number of Levels
Number of patches	Total number of food patches on the landscape	250 - 3500	82
Patch clustering	Degree of spatial clustering of patches (μ parameter controlling aggregation)	0 - 30	59

Scenario Component	Parameter Description	Values Tested	Number of Levels
Patch calorie variance	Calorie Coefficient of Variance	0 - 5.25	31

5.2 Animals

Individuals are initialized by the assignment of a body mass in grams. These individuals are considered to be of a theoretical, generalized mammal, herbivorous species. There is no stochasticity in the assignment of body mass. The number of generations to be run, the number of repetitions desired, and the number of individuals simulated within each replicate is also chosen at this time. Further, the sampling interval and duration of an individual's "life" is determined using two separate allometric relationships. The lifespan is reduced by a factor of 1000 to maintain computational feasibility while preserving relative energetic and movement relationships.

Table 3: Allometric scaling equations for movement parameters. Note that body mass, M , is needed in grams (g).

Parameter	Equation	Source
Field Metabolic Rate	$10^{0.774} \times M^{0.727}$	Nagy, 1987
Movement costs	$10.7 \times M^{-0.316} \times velocity + 6.03 \times (\frac{M}{1000})^{-0.303}$	Taylor et al., 1982
Lifespan	$4.88 \times M^{0.153}$	Magalhães et al., 2007
95% Home Range	$HR_{95} = 10^{0.5079} \times M^{1.3721}$	Noonan et al., 2020
Variance in position	$\sigma_p^2 = \frac{HR_{95}}{-2 \ln(0.05)\pi}$	Noonan et al., 2020
Position autocorrelation	$\tau_p = 10^{1.2994} \times M^{0.8129}$	Noonan et al., 2020
Position autocorrelation	$\tau_v = 10^{-1.3652} \times M^{0.7871}$	Noonan et al., 2020
Offspring requirements	$0.02425 \times M^{0.92}$	Blueweiss et al., 1978; Young et al., 2021

6 Input data

The model does not use input data to represent time-varying processes. Empirical allometric relationships are built into the model to calculate ecological relationships.

7 Submodels

7.1 Landscape Modelling

To create habitats which individuals could gain resources from, we utilized the `rMatClust()` function from the `spatstat.random` package in R. This function generates a realisation of a Matérn cluster process, within a specific window.

The extent of the landscape is defined as a function of the positional variance in an animal's movement, such that

$$EXT = \sqrt{(-2 \times \log_{10}(0.001) \times \pi) \times \sigma} \quad (1)$$

This window extents past the 95% home range area, and ensures that simulated tracks do not leave the landscape.

To ensure uniformity across simulations, we calculated the intensity of the landscape, λ , as the ratio of the number of desired points and the extent of the simulated landscape. The parent intensity, κ , was defined as $\kappa = \lambda/\mu$, where μ is the expected number of offspring points per parent point. We kept the radius around each parent point constant ($r = 1$) for consistency. This setup allows for the adjustment of minimal parameters to adjust the distribution and density of the landscape to mimic a wide variety of ecological scenarios.

To decrease computational time, the habitat is generated as 500 "tiles", each using the inputted model parameters. These tiles are then stitched together to create a single habitat which is used in subsequent simulation steps.

The landscapes were then thinned using the `rthin()` and `sample.int()` functions, in

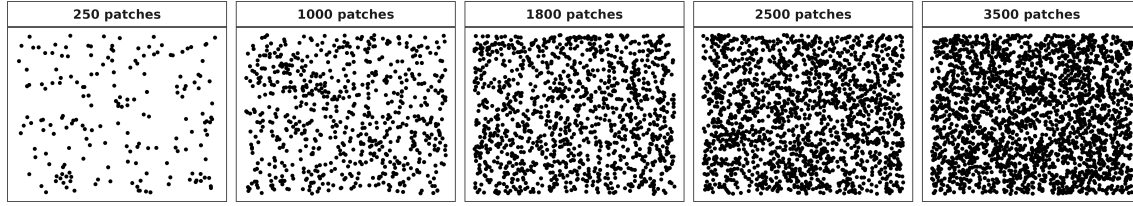


Figure 1: Examples of landscape density tested through simulation.

necessary, to reduce the number of points to the exact desired value.

`rthin()` applies independent random thinning, using the probability $p = n_{points}/1.01$ to reduce the total number of points slightly above the desired value. This is chosen to ensure the plot is not over-thinned by the stochastic thinning process.

`sample.int()` then removes the final points needed to reach the targeted number of patches exactly.

7.1.1 Landscape Density

The landscape framework allows for flexible landscape creation of customized landscapes with the alteration of single parameters. To test the effect of resource density, we increased the parameter `n_points`. See Figure 1 for examples.

7.1.2 Landscape Clustering

To evaluate the change in movement strategy as a result of resource spatial distribution, we ran simulations along a gradient of resource clustering. We started with a homogeneous Poisson point process and increased clustering following a Matérn clustering process. We increased the value of parameter `mu` in order to increase the degree of clustering (Figure 2).

7.1.3 Patch Unpredictability

To look at the effect of resource unpredictability, we increased the coefficient of variance for the caloric value of patches in order to simulate environmental unpredictability. See Figure 3 for implemented examples.

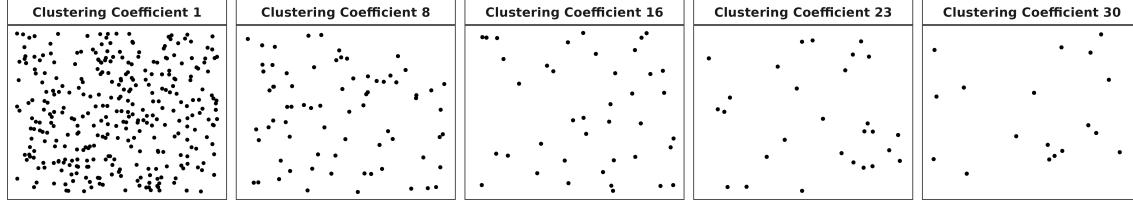


Figure 2: Examples of landscape clustering tested through simulation. Clustering coefficient refers to the value of μ used to create the landscape.

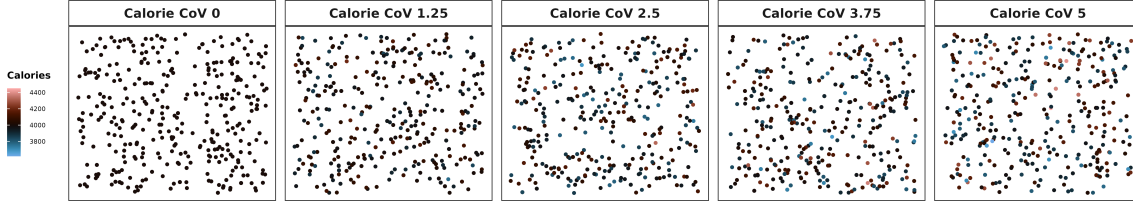


Figure 3: Examples of landscape unpredictability tested through simulation. Caloric CoV refers to the coefficient of variance added to the caloric value of patches.

7.2 Movement Centres

Individual starting locations were generated using the `rbvpois()` function from the `extraDistr` package in R, which draws samples from a bivariate Poisson distribution. This distribution was used to generate spatial coordinates for the initial centres of individuals. The parameters of the distribution were set as functions of the spatial variance parameter (σ_p), ensuring that the starting locations were distributed within the expected spatial extent of the individual's home range.

7.3 Movement Parameters

For the first generation of each simulation, allometric relationships derived from Noonan et al., 2020 were used to generate the initial values of τ_p , τ_v , and σ using the initial population wide body mass (see Table 3). The initial population had values of τ_v and τ_p which spanned from 0 to twice the allometrically calculated value, following an uniform distribution. The value of σ remained constant for all individuals for the duration of the simulation.

Individuals were simulated for a portion of their lifespan ($1/1000$), defined via the allometric relationship described in Magalhães et al., 2007, such that $t = 4.88 * M^{0.153} * 0.001$

which was then converted to seconds. The sampling interval for each simulation was defined as a fraction of the velocity autocorrelation timescale, defined as $dt = \tau_v/10$, based on sensitivity analyses. These analyses revealed that this sampling interval was the largest possible while ensuring that individuals hit the patches along their track of movement. By keeping this interval as large as possible, we reduce computation time.

7.4 Movement Simulation

We used Ornstein-Uhlenbeck with Foraging (OUF) stochastic processes to simulate the movement of a founding population of prey individuals, using the `ctmm` package in R (Fleming et al., 2014). The model incorporates range residency behaviour with a stationary variance defined by $var[p]$ (in m^2), correlated positions with fixed autocorrelation timescale (τ_p , in sec), and correlated velocities with fixed autocorrelation timescale (τ_v , in sec). The movement model is well established and grounded in empirical data, providing realistic insights into mammalian movement (Noonan et al., 2019).

7.5 Foraging

Foraging success is calculated after the full movement trajectory is simulated, via a nearest neighbour analysis using the `RANN` package. For each animal entity coordinate along the movement track, the nearest landscape entity point (i.e., a resource patch) is calculated within a radius equal to the perceptual range of the animal entity. This perceptual range is defined as a function of the positional variance, given by $pr = \sigma \times 0.05$. In cases where there is more than one point within this radius, the closest one is consumed. Once a patch has been consumed, it is removed from the landscape and the individual cannot consume it again. Once a patch has been consumed, or if there are no patches to consume at that time step, the algorithm moves to the next position of the movement trajectory. This process is repeated until the end of the movement track (Figure 4). After the consumed patches are identified, total caloric gains are calculated. Caloric value of a patch is stored as a 'mark' in the point process and allows for the calculation of total energy gains to be straightforward.

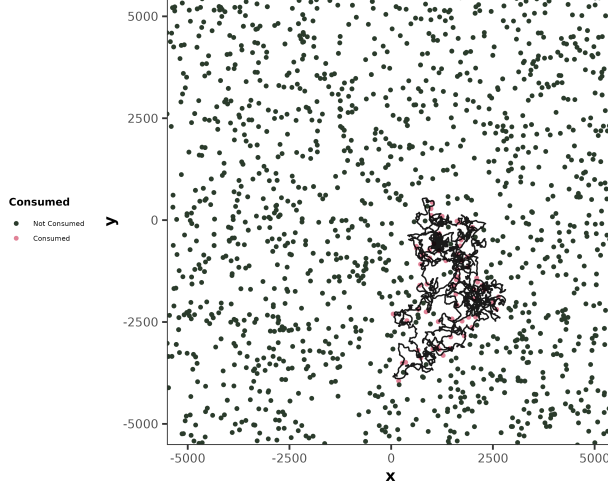


Figure 4: Example landscape and movement track, with patches coloured by consumption status. Green patches are those not consumed by the foraging model, while pink patches are.

For an individual, the total energy gained is equal to the sum of all the caloric values for all patches encountered.

7.6 Energetics

Following the calculation of gross caloric gains, the model proceeds to account for energetic costs that animals face. This is composed to two components, i) the field metabolic rate, and ii) the energetic cost of movement, such that total energy available for growth and reproduction is equal to $E_{\text{net}} = E_{\text{foraging}} - E_{\text{metabolic-total}} - E_{\text{transport-total}}$ where E_{foraging} is the total calories gained from movement, $E_{\text{metabolic}}$ is the field metabolic rate, and $E_{\text{transport}}$ is the cost of movement.

The Field Metabolic Rate is the estimated cost of all energy expenditures, including the basal metabolic rate, thermoregulation, and digestion (Nagy, 1987). By using this metric, we capture many of the energy expenditures that organisms face. We use the allometric

equation for herbivorous mammals derived from Nagy, 1987, given by

$$\log_{10}(E_{\text{metabolic}}) = 0.774 + 0.727 \log_{10}(M) \quad (2)$$

We convert this from *kJ/day* to *kcal/sec* and then calculate the total metabolic expenditure as the product of this rate and the total simulation time, such that

$$E_{\text{metabolic-total}} = E_{\text{metabolic-rate}} \left(\frac{\text{kcal}}{\text{sec}} \right) \times \text{duration}(\text{sec}) \quad (3)$$

The energetic cost of movement was calculated according to Taylor et al., 1982. In this case, the cost of movement is a function of the body mass of an individual and its movement velocity (see Eq 4). The velocity factor of this is important, as it restricts overly fast movement due to high metabolic costs, something that is not commonly seen in this type of model. This integration allows us to understand how movement can optimize when infinite speed and diffusivity is not possible.

$$E_{\text{transport-rate}} = \frac{(10.7M^{-0.316} * v + 6.03M^{-0.303}) * M}{1000} \quad (4)$$

where v represents mean movement speed, in m/s. We then convert this rate from *watts/kg* to *kcal/sec* and calculate the total energy expenditure as the product of the rate and total simulation time, such that

$$E_{\text{transport-total}} = E_{\text{transport-rate}} \left(\frac{\text{kcal}}{\text{sec}} \right) \times \text{duration}(\text{sec}) \quad (5)$$

7.7 Reproduction

After subtracting energy expenditures from foraging gains, the remaining calories are available for growth and reproduction. To allocate available resources, we followed the DEBkiss framework, which is a KISS principle ("keep it simple, stupid") application of the Dynamic

Energy Budget theory (Jager et al., 2013).

Following generalised framework, we assumed 80 percent of excess resources are allocated to somatic maintenance and growth. We further assumed that it takes 5 kcalories to gain 1 gram of mass Waterlow, 1981. While this is an overestimation of the weight gained by an individual over time, it allows us to further restrict the ability for offspring production, as this updated body mass is used in subsequent scaling relationships which calculate offspring produced. The updated mass of an animal is calculated as

$$mass_{updated} = \frac{E_{net} * 0.8}{5} \quad (6)$$

For fitness calculation, we use the estimate described by Jager et al., 2013, where the cost of one offspring is

$$offspring = floor(\frac{W_R}{W_{B0}}) \quad (7)$$

where W_{B0} represents the mass of assimilates buffer in an egg. Since mammals do not lay eggs, we used the birth weight of a neonate as a proxy, based on the allometric relationship defined by Blueweiss et al., 1978 and the assumption that mammals are approximately 25% of a neonate's weight is comprised of dry assimilates Young et al., 2021. For this calculation, we used the 'updated' animal weight to calculate the mass of the neonate, such that

$$W_{B0} = 0.25 * (0.097 * mass_{updated}^{0.92}) \quad (8)$$

W_R represents the mass allocated to reproduction in kcal. This is based on the assumptions that 20 percent of gained resources are allocated towards reproduction and that 1

265 gram of body mass requires 5 kcalories, similar to how somatic maintenance and growth is
 266 calculated (Waterlow, 1981).

$$W_R = \frac{E_{net} * 0.2}{5} \quad (9)$$

267 7.8 Inheritance

268 At the end of each generation, all individuals with more than 0 offspring have their movement
 269 parameters (τ_p, τ_v) saved in preparation for the subsequent generation. New individuals in
 270 the next generation then receive their movement traits from this pool. Gaussian noise is
 271 added to mimic mutation based variance, using the `rnorm` function from the `stats` package,
 272 with a mean of zero and standard deviation of 10 or 2 for τ_p and τ_v respectively. On the
 273 other hand, σ is determined via allometry in all generations, thus constraining individuals
 274 to navigate an equally sized home range. By sampling new individuals from a pool of
 275 parental traits, we keep the population size constant, preventing infinite growth and high
 276 computational costs.

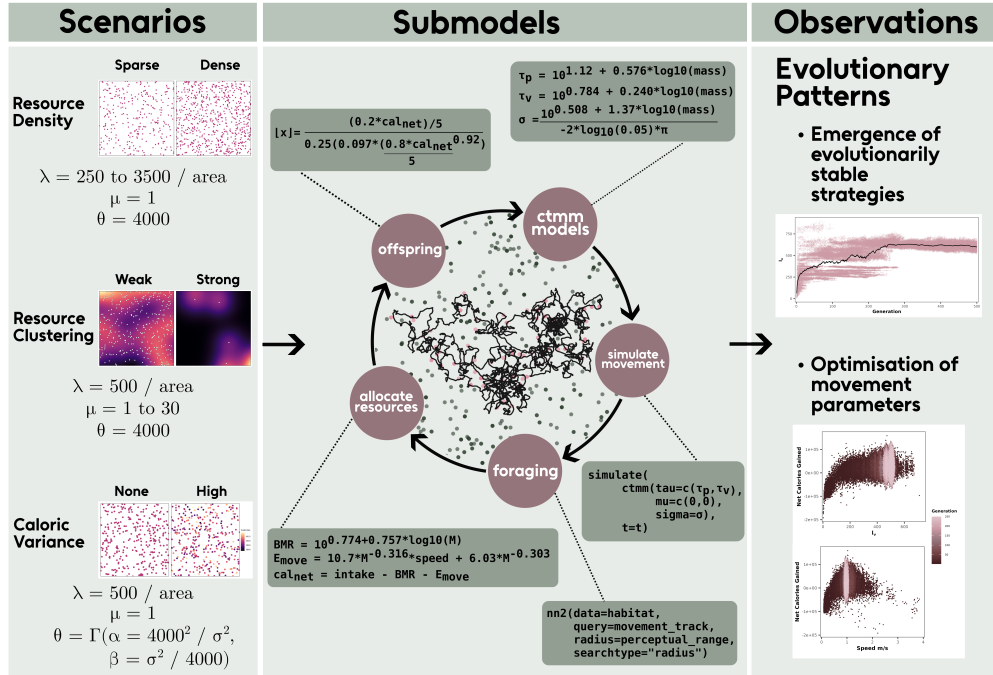


Figure 5: Visual overview of the model. During initialization animal entities are informed by allometric scaling relationships and a landscape scenario is created. Upon starting the simulation, animal entities generate movement parameters and their movement trajectory is simulated. Animals then forage, allocate resources, and ultimately produce offspring. This is repeated until the simulation is terminated at 500 generations (See supplementary material A for further details).

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